FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE **JEE (Advanced), 2016**

HALF COURSE TEST – VI

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 240**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 04)** contains 4 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (05 – 08) contains 4 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

   Section-A (09 to 11) contains 1 paragraph having 3 questions. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct matching. No marks will be given for any wrong matching in any question. There is no negative marking.

3. **Section-C (01 – 06)** contains 6 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Single Correct Choice Type

*This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

1. A railway track is banked for a speed v, by making the height of the outer rail 'h' higher than that of the inner rail. The horizontal separation between the rails is d. The radius of curvature of the track is 'r' : then which of the following relation is true?

(A) $\frac{h}{d}=\frac{v^2}{rg}$ (B) $\tan\left(\sin^{-1}\frac{h}{d}\right)=\frac{v^2}{rg}$

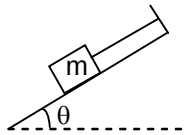
(C) $\tan^{-1}\left(\frac{h}{d}\right)=\frac{v^2}{rg}$ (D) $\frac{h}{r}=\frac{v^2}{dg}$

2. A horizontal spring mass system of mass M performs simple harmonic motion of amplitude a_0 and time period T_0. When the mass M passes through mean position, another stationary mass M sticks to it and both move together. If a and T be new amplitude and time period, then

(A) $a=\frac{a_0}{\sqrt{2}}$ (B) $a=a_0\sqrt{2}$

(C) $T=\frac{T_0}{\sqrt{2}}$ (D) None

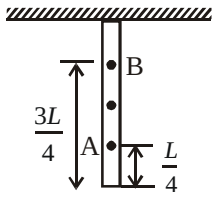
3. A metal block of mass m is placed on a smooth metallic plane in support with string as shown in diagram. If, after long time due to corrosion, the contact surface becomes rough with coefficient of friction μ, then friction force acting on the block will be



(A) $\mu mg\cos\theta$ (B) $mg\sin\theta$

(C) mg (D) None

4. A string of mass *m* and length *l* is hanging from ceiling as shown in the figure. Wave in string moves upward. v_A and v_B are the speed of wave at *A* and *B* respectively. Then v_B is



(A) $\sqrt{3}v_A$ (B) v_A

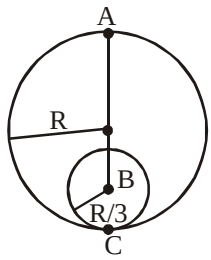
(C) $< v_A$ (D) $\sqrt{2}v_A$

Space for rough work

Straight Objective Type

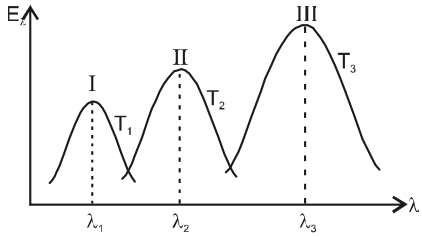
*This section contains **4 multiple** choice questions numbered 5 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE or MORE** are correct.*

5. Inside a uniform sphere of mass M and radius R, a cavity of radius R/3 is made in the sphere as shown.



(A) Gravitational field inside the cavity is uniform.
(B) Gravitational field inside the cavity is non–uniform.
(C) The escape velocity of a particle projected from point A is $\sqrt{\frac{88GM}{45R}}$.
(D) Escape velocity is defined for earth and particle system only.

6. Three identical bodies are at temperature T_1, T_2 and T_3 having e_1, e_2 and e_3 as their respective emissivities. The thermal spectrum obtained for them is as shown in the diagram. Choose the correct order of temperatures and emissivities:



(A) $T_1 > T_2 > T_3$
(B) $T_1 < T_2 < T_3$
(C) $e_1 < e_2 < e_3$
(D) $e_1 > e_2 > e_3$

7. An ideal monoatomic gas undergoes a process in which $P = 3V^3$. Choose the correct statements.
(A) If volume increases then temperature will also increase
(B) If temperature of the gas decreases, then pressure must also decrease.
(C) For expansion, heat will have to be supplied to the gas.
(D) If temperature increases then work done by gas is positive.

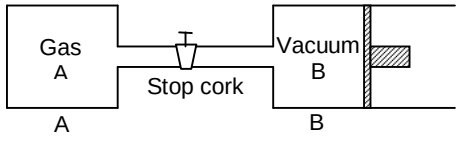
8. Which of the following will have a different time period (compared to that on the earth), if taken to the moon?
(A) A simple pendulum.
(B) A spring mass system oscillating vertically in the gravitational field.
(C) A torsion pendulum.
(D) An hourglass clock, which is essentially a vessel filled with a liquid, emptying through a hole in the bottom.

Space for rough work

Comprehension Type

*This section contains one group of questions. This group has 3 multiple choice question based on a paragraph. This question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct*

Paragraph for Question Nos. 9 to 11



An ideal diatomic gas is confined in a cylinder A of volume V_0. Then, cylinder is connected to another cylinder B with the help of tube of a negligible volume. The cylinder B is fitted with a movable piston, which can be adjusted from outside. Initially, piston is adjusted so that the volume of B is same as volume of A, i.e., V_0. B is evacuated and stop cork is opened so that gas expands and occupies the volume $2V_0$.

Answer the following questions based on above information.

9 During this free expansion, the internal energy of this system
(A) Increases (B) Decreases
(C) Remains constant (D) Nothing can be said

10 With open stop cork, the piston is slowly moved to compress the gas back to cylinder A at temperature T. Then, for n moles of gas, work done on the gas is
(A) $nRT\ln 2$ (B) $-nRT\ln 2$
(C) nRT (D) $-nRT$

11 The heat absorbed by the gas in the above process is
(A) $nRT\ln 2$ (B) $-nRT\ln 2$
(C) nRT (D) $-nRT$

Space for rough work

SECTION – B

(Matching List Type)

*This section contains **2 questions.** Each question has four statements (A, B, C and D) given in **Column I** and **five statements** (p, q, r, s and t) in **Column II.** Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given q and r, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS.*

1. In Column I, some statements or expressions related to first law of thermodynamics are given and corresponding processes are given in Column-II. Match the entries of Column I with that of Column-II:

Column I		Column II	
(A)	Work done by the system on surroundings can be non-zero in	(p)	Adiabatic process
(B)	$dU = nC_v dT$ is valid for	(q)	Isothermal process
(C)	dU is zero for	(r)	Isothermal expansion process
(D)	$dQ = nCdT$ is non-zero for	(s)	Polytropic process
		(t)	Isochoric process

2. Match the following:

Column A		Column B	
(A)	Phase difference between any two particles can have any value between O to 2π.	(p)	Stationary waves.
(B)	Energy is transferred from one place to an other place.	(q)	Plane simple harmonic transverse waves.
(C)	Phase difference between any two particles is either zero or π.	(r)	Sound waves.
(D)	Amplitude of vibration of all particles are equal.	(s)	Standing waves in an open organ pipe.
		(t)	Plane simple harmonic longitudinal waves.

Space for rough work

SECTION – C

Integer Answer Type

*This section contains **6** questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The bubble corresponding to the correct is to be darkened in the ORS.*

1. A string of length 2L, obeying Hooke's Law, is stretched so that its extension is L. The speed of the transverse wave travelling on the string is v. If the string is further stretched so that the extension in the string becomes 4L. The speed of transverse wave travelling on the string will be $x\sqrt{2}\,v$. Calculate the value of x.

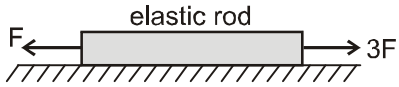
2. A train moves towards a stationary observer with a speed of 34 ms^{-1}. The train sounds a whistle and its frequency registered by the observer is f_1. If the train's speed is reduced to 17 ms^{-1}, the frequency registered is f_2. If the speed of sound is 340 ms^{-1}, then the ratio $\frac{f_1}{f_2}$ is $\frac{10+x}{18}$. Calculate the value of x.

3. A tuning fork A whose frequency is given by the manufacturer as 512 Hz. It is being tested using an accurate oscillator. It is found that they produce 2 beats per second when the oscillator reads 514 Hz and 6 beats per second when it reads 510 Hz. The actual frequency of the fork in Hz is (510 + x). Calculate the value of x.

Space for rough work

4. A small block of mass 1 kg is moving with constant speed of 10 m/s on a typical path in a x – y vertical plane whose equation is $y = \frac{x^3}{30} - \frac{x^2}{10}$. The coefficient of friction between the block and path is 0.01. Find the magnitude of power dissipated by frictional force in watt at x = 2 m?

5. A uniform elastic rod of cross- sectional area A, natural length L and Young's modulus Y is placed on a smooth horizontal surface. Now two horizontal forces (of magnitude F and 3F) directed along the length of rod and in opposite directions act at its ends as shown. After the rod has acquired steady state, the extension of the rod is given by $\frac{xF}{YA}L$. Calculate x.



6. A uniform square plate of mass m = 100 gm and side a = 24 cm can freely rotate about a vertical axis passing through one edge. It is initially at rest. A particle of mass m = 100 gm is moving horizontally and perpendicular to the plane of the plate with velocity u = 70 cm/s. The particle collides with the plate elastically at the centre of the plate. Find the angular velocity (in rad/s) of the plate just after collision.

Space for rough work

Chemistry

PART - II

SECTION – A

Single Correct Choice Type

*This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

1. Which of the following is paramagnetic?
   (A) O_2^- (B) CN^-
   (C) CO (D) NO^+

2. Which of the following species is incorrect matched with its shape?
   (A) $ClF_3 \longrightarrow T-$shaped (B) $SO_3^{2-} \longrightarrow$ tetrahedral
   (C) $C_2O_4^{2-} \longrightarrow$ planar (D) $I_3^- \longrightarrow$ linear

3. Identify the incorrect statement (s) among the following:
   (A) if spin quantum number curve having 3 possible values then 4th period of periodic table would have maximum of 27 elements.
   (B) platinum (Pt) is one of the typical transition metal.
   (C) total number of f – electrons in Gd (at. No. 64) is 8.
   (D) atomic radius of Ga is nearly equal to that of Al.

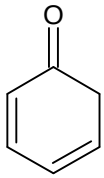
4. SO_3(g) is heated in a closed vessel. A equilibrium $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$ is established. The vapour density of the mixture, in which SO_3 is 50% dissociated is?
   (A) 32 (B) 35
   (C) 42 (D) 55

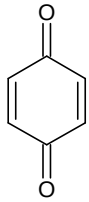
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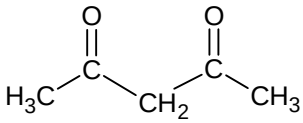
Multiple Correct Answers Type

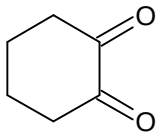
*This section contains 4 multiple correct answer(s) type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE** is/are correct.*

5. Which of the following compounds exhibit keto – enol tautomerism?

(A) 

(B) 

(C) 

(D) 

6. Which of the following carbide on reaction with water give methane?
   (A) CaC_2 (B) Be_2C
   (C) Mg_2C_3 (D) Al_4C_3

7. Which of the following statement is/are true regarding a real gas?
   (A) Boyle temperature is the temperature at which the slop $\left(\frac{dz}{dp}\right)$ of the curve between compressibility factor vs P is same as that for an ideal gas over a wide range of pressure.
   (B) At larger molar volumes, a real gas always obeys ideal gas laws.
   (C) At low pressure, attractive forces in the real gases dominates and the gas is more compressible.
   (D) At critical state, the compressibility factor is 3/8.

8. Which of the following reaction is/are correct?
   (A) $3CS_2 + 6NaOH \longrightarrow Na_2CO_3 + 2Na_2CS_3 + 3H_2O$
   (B) $SiCl_4 + LiAlH_4 \longrightarrow SiH_4 + AlCl_3 + LiCl$
   (C) $SiO_2 + 4HF(\text{excess}) \longrightarrow SiF_4 + 2H_2O$
   (D) $Na_2B_4O_7 + H_2SO_4 \longrightarrow Na_2SO_4 + H_3BO_3$

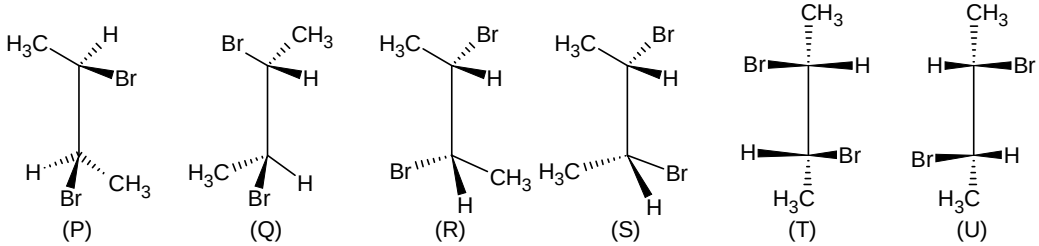
Space for rough work

Linked Comprehension Type

*This section contains **one group** of questions. This group has 3 multiple choice question based on a paragraph. This question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct*

Paragraph for question Nos 9 to 11

A structure with two stereogenic carbon does not always have four possible stereoisomers. Sometimes are only three. This happens because some molecules are achiral even though they contain stereogenic carbon atoms. Let us write stereo chemical formula for 2, 3 – dibromo butane.



Observe the above configurations carefully and answer the following questions:

9. Which of the following pair represents the identical compounds?
   (A) P & T (B) Q & T
   (C) R & S (D) S & T

10. Consider the following pairs:
   I. Q & S
   II. P & R
   III. T & R
   IV. P & U
   Which of the above pairs represents enantiomers?
   (A) I, II, III (B) II, III, IV
   (C) II & III (D) I, II, III & IV

11. Which of the following configuration is chiral?
   (A) Q (B) S
   (C) R (D) all of these

Space for rough work

SECTION – B
(Matrix-Match Type)

*This section contains **2 questions.** Each question has four statements (A, B, C and D) given in **Column I** and **five statements** (p, q, r, s and t) in **Column II.** Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given q and r, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS.*

1. Match the statement in column – I with corresponding type (s) of silicates in column - II

Column – I		Column – II	
(A)	Only one corner of SiO_4^{4-} units is shared	(p)	Chain silicates
(B)	At lest one corner of SiO_4^{4-} unit is shared	(q)	Sheet silicates
(C)	Only two corners of SiO_4^{4-} units are shared	(r)	Pyrosilicates
(D)	At least two corners of SiO_4^{4-} units are shared	(s)	Cyclic silicates
		(t)	sp^3 hybridised Si – centre

2. Match the compounds in column – I with their corresponding decreasing K_a value in column – II.

Column – I (compounds)		Column – II (Acidity)	
(A)	o, m, p – nitrophenols	(p)	o > m > p
(B)	o, m, p – cresols	(q)	p > o > m
(C)	o, m, p – chlorophenols	(r)	p > m > o
(D)	o, m, p – nitrobenzoic acid	(s)	m > p > o
		(t)	All three are more acidic than water

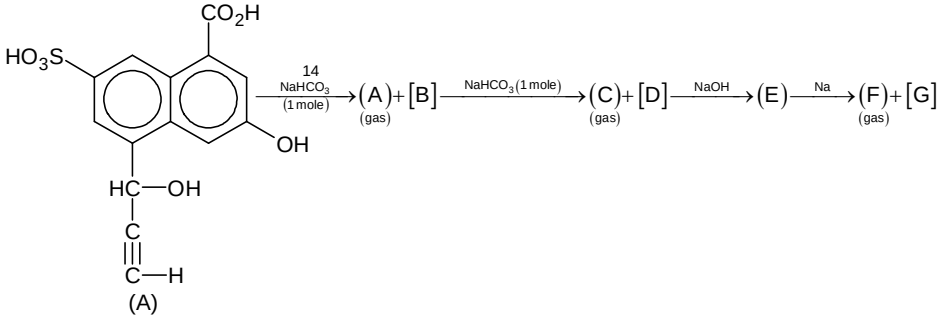
Space for rough work

SECTION – C

(Integer Answer Type)

*This section contains **6** questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The bubble corresponding to the correct is to be darkened in the ORS.*

1. Consider the following reaction sequence of compound (A).



Find the total sum of different types of atoms present in gases A, C and F?

2. Haemoglobin contains 0.25% iron by weight. The molecular weight of haemoglobin is 89600. What is the number of iron atom per molecule of haemoglobin? (atomic weight Fe = 56)

3. At 400 K, the rms speed of a gas x (MW = 40) is equal to most probable speed of gas Y at 60 K. The molecular weight of Y is.

4. K_{sp} of $Cu(OH)_2$ is 5×10^{-16} at 25°C, then pH of a saturated solution of $Cu(OH)_2$ is.

5. In a closed flask at 400 K solid $(NH_4)_2S$ was taken. After some time a constant pressure of 3 bar was found in the flask. K_p of the reaction:
$(NH_4)_2S(s) \rightleftharpoons 2NH_3(g) + H_2S(g)$ at 400 K is.

6. In a hydrogen atom an electron is in the 5th excited state. Total number of electronic transtions possible which will give vissible radiations.

Space for rough work

Mathematics PART – III

SECTION – A
Single Correct Choice Type

*This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

1. If $a_1, a_2, a_3 \ldots$ are positive numbers in G.P. and $x_1, x_2, x_3 \ldots$ are positive numbers such that $a_1^{\frac{1}{x_1}} = a_2^{\frac{1}{x_2}} = a_3^{\frac{1}{x_3}} = \ldots$ then $x_1, x_2, x_3 \ldots$ are in
   (A) A.P. (B) G.P.
   (C) H.P. (D) none.

2. A tangent drawn to hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at a point with eccentric angle $\frac{\pi}{6}$ forms a triangle of area $3a^2$ square units with coordinate axes, then the square of its eccentricity is equal to
   (A) 17 (B) 36
   (C) 28 (D) 4.

3. If each of the quadratic equation $x^2 + 2ax + b = 0$, $x^2 + 2bx + c = 0$ and $x^2 + 2cx + a = 0$ gas real and equal roots and $a \neq b \neq c$ and $a > 0, b > 0, c > 0$, then
   (A) a, b, c can't be rational numbers (B) a, b, c can't be integers
   (C) a, b, c can't be irrational numbers (D) none of these.

4. In ΔABC, if $\sin 2A + \sin 2B + \sin 2C = 4 - 4 \cos A \cos B$, then triangle is
   (A) equilateral (B) right angled but not isosceles
   (C) isosceles but not right angled (D) isosceles and right angled.

Multiple Correct Choice Type

*This section contains **4 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct*

5. Let PQ be a chord of the parabola $y^2 = 4x$. A circle drawn with PQ as a diameter passes through the vertex V of the parabola. If $ar(\Delta PVQ) = 20$ unit^2 then the coordinates of P are
   (A) (16, 8) (B) (16, –8)
   (C) (–16, 8) (D) (–16, –8).

Space for rough work

6. Let z be a complex number satisfying equation $z^n = \bar{z}^m$, where n, m $\in$ N, then
(A) if n = m, then number of solution of equation will be finite
(B) if n = m, then number of solution of equation will in finite
(C) if n $\neq$ m, then number of solution of equation will be n + m
(D) if n $\neq$ m, then number of solution of equation will be n + m + 1.

7. If $\left(3\sqrt{3}+5\right)^9 = R$ and [R] denotes the greatest integer less then or equal to R, then
(A) [R] is divisible by 10
(B) [R] is divisible by 3
(C) [R] is divisible by 15
(D) [R] is an even number.

8. Maximum number of regions into which 10 lines can divide a plane, is
(A) 42
(B) 56
(C) $^8P_2$
(D) $^7P_2$.

Comprehension Type

*This section contains **one group** of questions. This group has 3 multiple choice question based on a paragraph. This question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct*

Paragraph for Question Nos. 9 to 11

Read the following write up carefully and answer the following questions:

Consider the quadratic equation $ax^2 - (2a - 1)x + a - 3 = 0$ and answer the following questions:

9. If both the roots are real and are of opposite sign, then complete set of values of a is
(A) (0, 3)
(B) $(-\infty, 0) \cup (3, \infty)$
(C) $\left\{\frac{1}{2}\right\}$
(D) none.

10. If a is given as $\frac{n^2+n}{2}$, where 'n' is national number, then both the roots are necessarily
(A) Integers
(B) Rational numbers
(C) Even integers
(D) None.

11. As per the data of pervious question both the roots will lie in the interval
(A) [–1, 2]
(B) [1, ∞)
(C) [1, 2]
(D) None.

Space for rough work

SECTION – B

(Matching List Type)

*This section contains **2 questions.** Each question has four statements (A, B, C and D) given in **Column I** and **five statements** (p, q, r, s and t) in **Column II.** Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given q and r, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS.*

1. Match the following Column-I with Column-II

Column – I		Column – II	
(A)	If the straight lines $y = a_1x + b$ and $y = a_2x + b$, $(a_1 \neq a_2)$ and $b \in R$ meet the coordinate axes in concyclic points, then	(p)	$a_1^2 + a_2^2 = 4$
(B)	If the chord of contact of the tangents drawn to $x^2 + y^2 = 4$ and $b \in R$ from any point on $x^2 + y^2 = a_1^2$, touches the circle $x^2 + y^2 = a_2^2, (a_1 \neq a_2)$, then	(q)	$a_1 + a_2 = 20$
(C)	If the circles $x^2 + y^2 + 2a_1x - 2a_2y - 1 = 0$, $(a_1 \neq a_2)$ cut orthogonally, then	(r)	$a_1a_2 = 4$
(D)	If the shortest and largest distance from the point (8, 5) to the circle $x^2 + y^2 - 4x + 6y - 12 = 0$ are a_1 and a_2 respectively, then	(s)	a_1a_2
		(t)	$a_2 + a_1 = 10$

2. Match the following Column-I with Column-II

Column – I		Column – II	
(A)	The number of real roots of the equation $\cos^7 x + \sin^4 x = 1$ in $(-\pi, \pi)$ is	(p)	1
(B)	The value of $\sqrt{3}$ cosec 20º – sec 20º is	(q)	4
(C)	4 cos 36º – 4 cos 72º + 4 cos 36º sin 18º equals	(r)	3
(D)	The number of values of x where $x \in [-2\pi, 2\pi]$, which satisfy cosec x = 1 + cot x	(s)	2
		(t)	5

Space for rough work

SECTION – C

Integer Answer Type

*This section contains **6** questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The bubble corresponding to the correct is to be darkened in the ORS.*

1. If the equation $x^3 - bx^2y - cxy^2 + y^3 = 0$ represents three straight lines of which two are mutually perpendicular, then the value of (b + c) is

2. If $S_n = {}^nC_0\ {}^nC_1 + {}^nC_1\ {}^nC_2 + \ldots + {}^nC_{n-1}\ {}^nC_n$ and if $\frac{S_{n+1}}{S_n} = \frac{15}{4}$, then the sum of two possible values of n is

3. Normal chord to the parabola $y^2 = 8x$ at the point P(2, 4) and Q meets the curve again in R. If circumcenter of ΔPQR is (h, k) then h + k is

4. Number of all five digit numbers of the form 34x5y (x and y are digits) divisible by 36, is

5. An ellipse of major axis 4 and minor axis 3 is drawn tangential to the lines $x - 3y + 2 = 0$ and $3x + y - 4 = 0$. Centre of this ellipse will always lies on the circle, then the radius of the circle is

6. Let the lengths of the altitudes drawn from the vertices of a ΔABC to the opposite sides are 2, 2 and 3. If the area of ΔABC is Δ, then find the value of $2\sqrt{2}\Delta$.

Space for rough work